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SPECTROSCOPY

Statistical Characterization of the Phytochemical Characteristics of Edible Mushroom Extracts

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ABSTRACT

The antioxidant activity, total phenolic content, and total flavonoid content of 10 edible mushrooms species, including cultivated (*Pleurotus ostreatus, Agaricus bisporus* white and brown) and wild (*P. ostreatus, Macrolepiota procera, Cantharellus cibarius, Russula vesca, Russula alutacea, Boletus edulis* and *Agaricus campestris*), were determined. The extraction was performed using water and 50% water–ethanol and the caps and stipe were investigated separately. Water was the most appropriate solvent for phenolic compounds regardless of the sample. In contrast, the presence of ethanol in the solvent increased the extraction of flavonoids for cultivated *P. ostreatus* (caps and stipe), and wild *R. alutacea, R. vesca, A. campestris, P. ostreatus* (only caps) and *C. cibarius, M. procera* (only stipe). Significant differences between the antioxidant activities of the samples were registered in relationship with the different solvents. The antioxidant activity of water extract of dried *A. bisporus* brown (cultivated) showed the strongest 1,1-diphenyl-2-picrylhydrazyl radicals-scavenging assay (88.64%), while the *B. edulis* hydroalcoholic extract contained 74.93%. A detailed investigation into the functional group of phenolics and other organic compounds responsible with the antioxidant activity has been performed using Fourier transform infrared spectroscopy. The infrared spectra showed that the solvent-type directly influences the extraction process and, hence, the antioxidant activity. The present study contributes to information concerning mushrooms as sources of biologically active compounds. To investigate the correlations between phytochemical characteristics (i.e., phenolics, flavonoids and antioxidant activity) closely related to nature of solvents, the statistical analysis was performed.

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Antioxidant activity; edible mushrooms; extract; Fourier transform infrared spectroscopy; statistical approach; total phenolics

Introduction

From the oldest times, mushrooms have been consumed by humans due to their taste, aroma, and texture, being part of the normal diet as well as a delicacy for some people. The interest in consumption of edible mushrooms (cultivated and wild) is increasing in

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many countries, especially in view of their nutritional value. Thus, the mushrooms are high in protein, carbohydrate, fiber, vitamins, and essential amino acids. From the functional perspective, mushrooms are low in fat and cholesterol (Mattila et al. 2000). Wild edible mushrooms have high levels of proteins, vitamins, iron, zinc, sodium, and other minerals, and are a popular food (Ouzouni et al. 2009).

Several edible mushrooms have been found, in addition to their nutritional profile, to be medically active in different therapies. The reason is linked to their richness in bioactive compounds, such as phenolic compounds, polyketides, terpenes, steroids (Kues and Liu 2000), beta-carotenes, and vitamins A and C (Soares et al. 2009). Bioactivities exhibited by mushrooms include antimicrobial (Ofodile et al. 2005; Barros et al. 2007; Kitzberger et al. 2007) and antiviral activities (Zahid et al. 2006). According to Ruiz-Rodriguez, Santoyo, and Soler-Rivas (2009), mushrooms are phyllogenetically far from plants, with which they share few biochemical and metabolic similarities, being a potential source of new bioactive compounds. Corrêa et al. (2016) pointed out that the accumulation of some secondary metabolites (some having biological activities) represents the particular characteristic of growth and development of mushrooms in nature.

Scientific studies (Ruiz-Rodriguez, Santoyo, and Soler-Rivas 2009; Corrêa et al. 2016) confirmed recently that bioactive compounds from many edible mushrooms are involved in lowering the cholesterol levels, protect against various disorders including tumors, properties directly or indirectly related to their high antioxidant activity. Some medicinal mushrooms show to have cardiovascular, anticancer, antiviral, antibacterial, antiparasitic, anti-inflammatory, and antidiabetic properties (Kosanić et al. 2016).

Antioxidant properties are assigned to edible mushrooms, having in view their content in different bioactive compounds. From many reasons, mushrooms are considered to be a natural and good source of antioxidants (Sławińska, Radzki, and Kalbarczyk 2013; Boonsong, Klaypradit, and Wilaipun 2016) which play an important role for chemoprevention of diseases caused by reactive oxygen species (Kamra and Bhatt 2012).

Reactive oxygen species generated during oxidative phosphorylation by dihydronicotinamide-adenine dinucleotide phosphate oxidase are normal components of healthy cells and also mediators of the first defensive actions of cells (Liu et al. 2013). However, the recent work (Jeong et al. 2012) highlights that overproduction of reactive oxygen species and oxygen-derived free radicals may induce many diseases, such as rheumatoid arthritis, atherosclerosis, diabetes, cancer, and aging.

Studies on the antioxidant activity of edible mushrooms (Jeong et al. 2012; Liu et al. 2013; Mishra et al. 2013) reported different results not only because the content of bioactive compounds may vary considerably depending on the strain, but also due to the factors that affect their concentration: cultivation, developmental stage, age of the fresh mushrooms, storage conditions, and the extraction method. However, it seems that the bioactive compounds exhibiting antioxidant activity are represented by minerals (e.g., selenium, zinc), polysaccharide–protein complexes, ergothioneine, vitamins, flavonoids, and related polyphenols that are rare in mushrooms. Their list is not exhaustive, as they are not well defined yet in all species (Ruiz-Rodriguez, Santoyo, and Soler-Rivas 2009).

Other studies (Mau, Chao, and Wu 2001; Cheung and Cheung 2005) correlate the antioxidant activity of mushrooms with their total phenolic content, while others show contrary results. Thus, according to Ribeiro et al. (2006), *Suillus granulates* and *Suillus bellini* showed high levels of phenolic compounds, but these did not correlate with their antioxidant activity.

Fruit bodies and mycelia of several mushrooms have been reported to show antioxidant activity (Yang, Lin, and Mau 2002; Lee et al. 2008). Liu et al. (2013) found that the alcoholic extract of *Agaricus bisporus* had strong antioxidant activity. The main phenolic compounds determined were gallic acid, protocatechuic acid, catechin, caffeic acid, ferulic acid, and myricetin. The alcoholic extracts of *Coprinus comatus* were shown to be more effective in scavenging ability on hydroxyl radicals than hot water extracts. The naturally occurring antioxidant components included total phenols, tocopherols, flavonoids, and polysaccharides (Li et al. 2010). The maximum antioxidant activity was displayed by *Termitomyces heimii* and *Termitomyces mummiformis,* which showed highest amount of tannic, gallic, protocatacheuic, and gentisic acids (Puttaraju et al. 2006). *Agaricus bisporus* is a strain recognized for its relatively high antioxidant activity. It is supposed that the compounds responsible for this feature are cinnamic, *p*-hydroxybenzoic, protocatechuic, and caffeic acids (Mattila, Suonpää, and Piironen 2001) and phenolic amino acids (e.g., tyrosine, 3,4-dihydroxyphenilalanine, *γ*-L(+)-glutaminyl-4-hydroxybencene, *γ*-L(+)-3,4-dihydroxybencene (Soler-Rivas et al. 1998), perhaps together with ergothioneine (Ruiz-Rodriguez, Santoyo, and Soler-Rivas 2009)).

Phenolic compounds represent a large group of secondary plant metabolites which now attract great interest due to their benefits for human health (Radulescu et al. 2017). There are studies that have shown that the antioxidant activity exhibited by phenolic compounds offers protection against chronic-degenerative diseases, cardiovascular diseases, diabetes mellitus, and neurodegenerative diseases (Croft 1999; Jiménez et al. 2008; Martins et al. 2011). Apart from their ability to act as free radical scavengers, hydrogen donators, and singlet oxygen quenchers (Croft 1999), phenolic compounds improve the organoleptic properties of vegetable origin food and can be used for wide applications in food (e.g., natural preservatives against food degradation and development of functional food), and pharmaceutical products (Ballesteros et al. 2017). In addition, antihyperglycemic, antibacterial, antiallergenic, and anti-inflammatory are biological properties assigned to phenolic compounds (Martins et al. 2011; Liu et al. 2012; Mussatto 2015).

Flavonoids are naturally occurring phenolic compounds found in fruits, vegetables, and teas, characterized by a benzo-*y*-pyrone structure. They represent the most common and widely distributed group of plant phenolic compounds (Barros et al. 2008). The major classes include anthocyanidins, flavanols, flavanones, flavones, flavonols, isoflavones, and chalcones (Bakarm et al. 2009). Possessing strong antioxidant properties, flavonoids inhibit lipid peroxidation, and chelate ferrous ions. Due to their high pharmacological activities as radical scavengers, flavonoids are important for human health (Barros et al. 2008).

Several analytical techniques have been used to certify the quality of different vegetables and fungus extracts, including high-performance liquid chromatography (Radulescu and Hossu 2005; Radulescu, Hossu, and Ionita 2005), gas chromatography–mass spectroscopy (Li et al. 2017), Fourier transform infrared spectroscopy (Radulescu, Hossu, and Ionita 2006; Radulescu and Tarabasanu-Mihaila 2006; Radulescu et al. 2008; Barbes, Radulescu, and Stihi 2014; Bintintan et al. 2017), nuclear magnetic resonance (Radulescu, Hossu, and Ionita 2006; Radulescu et al. 2008), and inductively coupled plasma-mass spectroscopy (Georgescu et al. 2016).

Although there are many studies on edible and wild mushrooms, they report on species that are country-specific or are popular for consumption in a certain geographical area. Only few disparate results related to the 10 mushrooms species mentioned above can be found. Moreover, there are only few studies concerning the chemical composition of

mushroom caps and stipe, respectively; generally, the related measurements are provided for whole mushrooms. Previous work within our research group involved a systematic study about the level of several essential and nonessential metals correlated with nutritional and food safety aspects related to the consumption of traditional edible mushrooms from Dambovita County (Georgescu et al. 2016, 2017). The aim of this research was to achieve some correlations between phytochemical characteristics (i.e., antioxidant activity, total phenolic content, and total flavonoid content) in two extract types (i.e., aqueous and hydroalcoholic) obtained from 10 edible mushroom species (cultivated and wild) from Dambovita County. These traditional species are frequently used as human foods. The statistical evaluation highlights a clearly link between phenolic compounds and antioxidant activity of mushroom extracts (both caps and stipe) depending on solvent type.

Related to the authors' previous papers, the novelty of this paper consists in deepening research in the complete characterization of these traditional mushroom species through emphasizing their potential as food resources. In the context in which species showed their content of biological active compounds, future practical applications in the area of functional food will be developed by integrating the data concerning their lack of the toxicity and nutritional value. Related to other reports in the literature lack data focused on the characterization of mushroom species investigated in this paper. Moreover, the statistical analysis between total phenolics, flavonoids and antioxidant activity is less completely approached. This study represents a novel starting point for future studies in which these extracts can be used in different fields, such as medicine, pharmaceutics, cosmetics, and geriatrics.

Materials and methods

Materials and reagents

Ten edible mushroom species (six samples from each species, out of which seven were wild (*Pleurotus ostreatus, Agaricus campestris, Macrolepiota procera, Cantharellus cibarius, Russula vesca, Russula alutacea, Boletus edulis*) and three cultivated (*P. ostreatus, A. bisporus* white and *A. bisporus* brown) were collected during one harvesting season. All collected samples corresponded to the mature growing stage of the mushroom. The wild edible mushroom species were collected from several villages from Dambovita County, Romania, while a local producer from the same area provided cultivated species. The fresh edible mushrooms were cleaned, separated as cap and stipe samples, dried at 60°C to constant weight, and preserved in sealed containers. For analysis purposes, the dried samples were ground and aliquots from the obtained powder were weighed.

Reagents used for all experiments were of analytical grade from different suppliers: Folin–Ciocalteu reagent and 1,1-diphenyl-2-picrylhydrazyl (DPPH) from Merck, gallic acid, quercetin, solvents (e.g., ethanol 96%) and other chemicals (i.e., sodium nitrite, aluminum chloride, sodium hydroxide) from Sigma-Aldrich. Double distilled water (conductivity at 25°C bellow 0.5 $\mu S\,cm^{-1}$) was used for extractions, determination of phytochemical characteristics, and for other procedures applied during experiments.

Sample preparation

Cap and stipe extracts were distinctly prepared for each studied mushroom species at room temperature. The used solvents were double distilled water and 50% water–ethanol.

Weighed aliquots of 1 g were placed in glass vessels together with 25 mL solvent and stirred at 150 rpm for 4 h. The obtained extracts were centrifuged for 20 min at 2000 rpm in a Rotofix 46 Hettich centrifuge.

Methods

Antioxidant activity

Several chemical and biochemical assays can be used to screen the antioxidant properties of the mushrooms. However, the DPPH• assay is one of the most popular and frequently used method among the antioxidant assays (Akar, Küçük, and Doğan 2017). DPPH• is a stable, long-lived organic nitrogen radical with a strong absorption around 517 nm. The antioxidant compounds present in the extract release an electron or a hydrogen atom, converting thus the DPPH• to a more stable, diamagnetic molecule.

The antioxidant activity of the mushroom extracts was assessed using 2,2-diphenyl-1-picrylhydrazyl as in a method described in the literature (Mosquera, Correra, and Niño 2009). An aliquot of undiluted extract (0.75 mL for aqueous extracts and 0.5 mL for hydroalcoholic extracts) was mixed with 1.5 mL DPPH alcoholic solution (20 mg/mL). The samples were mixed and maintained at room temperature in dark for 30 min. The antioxidant activity was obtained using a double beam Evolution-260 BIO spectrophotometer (Thermo Scientific, USA).

For antioxidant activity determination, a solution prepared by mixing 0.75 mL of ethanol 96% and 1.5 mL of DPPH solution was used as the blank. The sample blank was prepared by mixing 0.75 mL extract with 1.5 mL of 96% ethanol. Each studied sample was tested in triplicate by the same procedure and the mean values were calculated, as well. The absorbance was measured for all samples at 517 nm. A lower absorbance indicates a higher radical-scavenging activity of the mushroom extract and vice versa. The results were depicted as the percent scavenging of 2,2-diphenyl-1-picrylhydrazyl at a fixed antioxidant concentration. The antioxidant activity (AA %) was calculated by

$$\mathrm{AA}(\%) = \left[A_{\text{reagent blank}} - \left(A_{\text{extract}} - A_{\text{sample blank}}\right)/A_{\text{reagent blank}}\right] \times 100 \qquad (1)$$

where $A_{\text{reagent blank}}$ is the absorbance of the reagent blank solution, A_{extract} is the absorbance of the sample with tested mushroom extract and DPPH solution, and $A_{\text{sample blank}}$ is the absorbance of sample blank solution. A low absorbance of the tested sample indicates a high free-radical-scavenging activity.

Total phenolics

The total phenolics in the mushroom extracts were determined with Folin–Ciocalteau reagent using the method described in ISO 14502-1:2005 (2005), as modified by Olteanu, Nicolescu, and Bumbac (2017) and Nicolescu, Olteanu, and Bumbac (2017).

Different dilutions of aqueous and hydroalcoholic extract samples were used. For aqueous extracts, the supernatant was first diluted (0.75 mL extract in double distillate water to 25 mL). Then 0.5 mL aliquots of this solution were used for spectrophotometric measurements. Briefly, to appropriate volumes of dilutions from each sample (i.e., 0.5 mL), 2.5 mL of 10% Folin–Ciocalteu reagent and 2 mL of 8% sodium carbonate were added. The mixtures were allowed for 60 min at room temperature in darkness. A similar algorithm was followed for the hydroalcoholic mushroom extracts, the difference in the sample initial dilution ratio (1.0 mL extract in double distillate water to 25 mL). The absorbance of all

samples was then measured at 765 nm using the double beam Evolution-260 BIO spectrophotometer (Thermo Scientific, USA). A calibration curve using gallic acid was plotted under the same conditions. The results are expressed as milligrams of gallic acid per gram of dry weight (mg GAE/g d.w.) of extract. All experiments were performed in triplicate and the mean ±standard deviation was reported.

Total flavonoid content

The total flavonoid content in aqueous and hydroalcoholic mushroom extracts was measured by the aluminum chloride colorimetric assay described by Kamtekar, Keer, and Patil (2014). An aliquot (i.e., 1 mL) of mushroom extract was added to 9 mL of double distilled water and 0.3 mL of 5% sodium nitrite solution were added. After 5 min, 0.3 mL of 10% aluminum chloride solution were added to the volumetric flask. After 5 min, 2 mL of 1 M sodium hydroxide were added. The volume was diluted to 10 mL with double distilled water and thoroughly mixed. The absorbance was measured at 510 nm against the blank (double distilled water). Samples were prepared in triplicate for each measurement and the mean absorbance value was calculated. The results were expressed as mg quercetin equivalents/mL, according to the calibration curve, and then converted to mushroom dry weight.

Attenuated total reflection–Fourier transform infrared spectroscopy

Attenuated total reflection–Fourier transform infrared spectroscopy allows the rapid, reagentless, inexpensive and high-throughput analysis of liquid and solid samples. To investigate the chemical functional groups of organic compounds in both mushroom extracts (i.e., aqueous and hydroalcoholic extracts of each cap and stem sample), Fourier transform infrared spectroscopy was used. The chemical changes induced by extraction techniques as well as the various functional groups responsible for biological activities were detected in the mid-infrared absorption region using a Vertex 80v spectrometer (Bruker) equipped with a diamond attenuated total reflection crystal accessory and a Hyperion infrared microscope. The extracts were placed on the sample chamber of attenuated total reflection–Fourier transform infrared spectrometer without any preparation. To ensure reproducibility, all samples were prepared in triplicate at room temperature (22°C) and presented as an average value. The important absorption frequencies were noted in the range of 3600–600 cm^{-1}, as well as the fingerprint region of the spectra.

Statistical analysis

The data analysis was performed using the Statistical Package for the Social Science 24.0 version software for MS Windows, to evaluate the analytical results and to investigate the correlations between parameters.

Results and discussion

2,2-Diphenyl-1-picrylhydrazyl radical-scavenging activity

The results of the DPPH•-scavenging activity of the ten edible mushrooms extracts (cap and stipe) are shown in Figure 1. They show that when water was used as solvent, *A. bisporus* brown (cap) presented the strongest DPPH•-scavenging activity (88.64%), while the smallest (25.72%) was determined for *M. procera* (cap). The values ranged between 74.93% (*B. edulis* - cap) and 13.61% (*R. alutacea* - stipe) using 50% water–ethanol

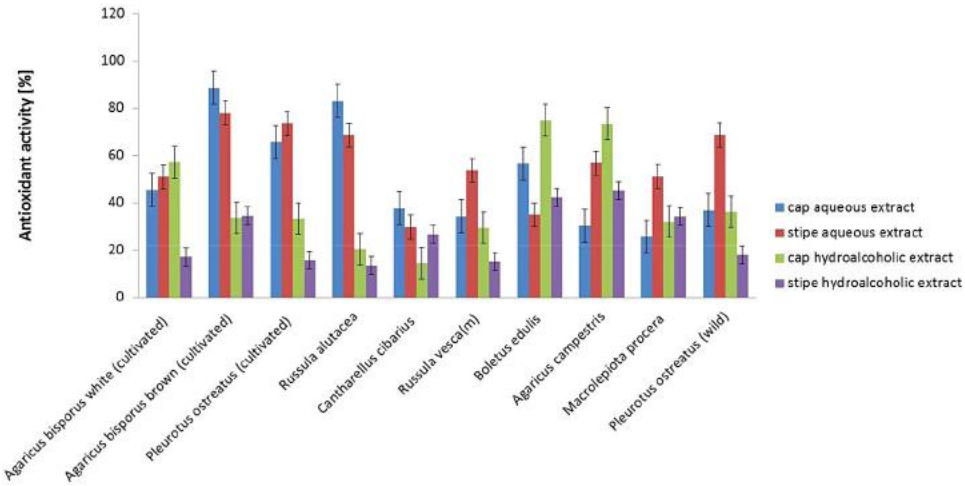

Figure 1. DPPH•-scavenging activity of extracts from 10 edible mushrooms (caps and stipe).

for extraction. The absorbance of the sample blank was between 0.003 and 0.018, while the absorbance of extract samples was in the range of 0.200–0.520.

In general, higher differences between results were registered having in view the mushroom species, their anatomic part, as well as the solvent type. Thus, the hydroalcoholic extracts obtained from the stipe of *C. cibarius, M. procera,* and *A. bisporus* brown contained a slightly higher antioxidant activity than the caps of the same species, in contrast to the other seven species analyzed. When water was used as solvent, the DPPH•-scavenging activity of extracts from mushrooms' stipe was also higher than in caps for *R. vesca, A. campestris, M. procera, P. ostreatus* (both cultivated and wild), and white *A. bisporus*. The differences were more obvious in this case.

The water extracts from analyzed mushrooms exhibited very good scavenging abilities on hydroxyl radicals. Thus, just five samples from the total of 20 (mushroom caps and stipe) registered higher antioxidant values than using 50% water–ethanol as the solvent by *B. edulis* (both caps and stipe), *A. campestris, M. procera,* and *A. bisporus* white (only caps). These results are in agreement only with the flavonoid content of the water–alcoholic extract from cap of *A. campestris*. Our results are in agreement with the results of Reis et al. (2012), who mentioned that *A. bisporus* (brown) was the species that presented the highest radical scavenging activity both fruiting body and mycelium.

This study had similar results to those reported by Ramirez-Anguiano et al. (2007), where the water extracts of *Pleurotus* sp., *A. bisporus,* and *B. edulis* showed higher antioxidant activities. Also, the ethanolic extract of *A. bisporus* from China had a strong DPPH•-scavenging activity (Liu et al. 2013). Moreover, Sławińska, Radzki, and Kalbarczyk (2013) reported that water extracts of *Flammulina velutipes* prepared by stirring at ambient temperature had higher scavenging activity toward the DPPH• (from $25.67 \pm 0.64\%$ to $56.28 \pm 0.49\%$) than extracts from other solvents (ethanol, methanol, and acetone). The results from Figure 1 related to the antioxidant activity of the mushrooms' ethanolic extracts are in agreement with the observations of Li et al. (2010), who pointed out that extracts from the caps showed better scavenging ability on 2,2-diphenyl-1-picrylhydrazyl than extracts from stipe of *C. comatus*.

The current study had different results to those reported by Jayakumar, Thomas, and Geraldine (2009), where the hydroalcoholic extract of *P. ostreatus* (prepared by extracting 5 g of mushroom dried with 100 ml of 95% ethanol, followed by solvent evaporation till 5.86% yield) exhibited the most potent radical-scavenging activity. Regardless, if *P. ostreatus* was cultivated or wild and of the anatomic part evaluated, the present study showed that the water was the most efficient solvent (especially for the stipe of the cultivated species) in relationship with the scavenging effects on DPPH•. In addition, Palacios et al. (2011) also reported that a methanolic extract with *A. bisporus* showed the lowest antioxidant activity (10% inhibition of lipid oxidation).

Ruiz-Rodriguez, Santoyo, and Soler-Rivas (2009) pointed out that the results concerning the antioxidant properties of many mushroom species varied from author to author, probably because of different environmental conditions, cultivation methodologies, developmental stage, and genetic variation within strains. In agreement with our results, the authors reported also that most commonly consumed mushrooms (*A. bisporus, P. ostreatus,* and *B. edulis*) were, generally, highlighted with higher antioxidant activities compared to the other species.

Moreover, the strong DPPH•-scavenging activity was shown in the mushroom extracts (both aqueous and hydroalcoholic) in which total phenolic and flavonoid contents were not the highest, suggesting this way that there are other compounds than phenolic ones responsible for their antioxidant activity. However, from the analyzed species, *B. edulis* registered high total phenolic and flavonoid contents in aqueous extracts (both for cap and stipe) which in turn lead to a higher antioxidant activity both in aqueous and 50% water–ethanol extracts (56.53% for cap and 34.98% for stipe).

Total phenolics in mushroom extracts

The total phenolic content of the 10 edible mushrooms (water and 50% water–ethanol extracts) is provided in Table 1. The results are expressed as milligrams of GAE per gram of dried weight (d.w.).

The results indicated relatively higher differences between the amounts of phenolic compounds found in mushroom extracts, the hydroalcoholic solvent was more efficient, regardless if caps or stipe were evaluated. Thus, the concentrations ranged between 9.28 ± 0.03 mg GAE/g d.w. (cap of *C. cibarius*) and 69.65 ± 0.23 mg GAE/g d.w. (cap of *A. campestris*) as average values for the cultivated species. Taking into account both solvents and the mushroom species, *A. campestris* (cap) registered the highest difference between the content of phenolic compounds in the sample prepared in ethanol instead of water, while the smallest was determined in the case of *M. procera* (stipe), which showed a slight preference for water from the above-mentioned point of view. Intermediate differences were established for the caps of *R. vesca, R. alutacea,* and *A. bisporus* white, for which the alcoholic environment was favorable to phenolic compound extraction, while for *B. edulis* (stipe), water was the most suitable to extract the polyphenols.

The phenolic compound content in water extracts was higher in caps than in stipe only for *P. ostreatus* (cultivated), *R. alutacea, B. edulis,* and *M. procera.* However, when the water–ethanol mixture was used, in *C. cibarius,* the total phenolics of mushrooms' were higher in stipe than in caps (by 4.93 times in the case of *A. campestris*). Relative closer values between the content of the phenolic compounds in caps and stipe have been

Table 1. Total phenolic content in the mushroom extracts. Results are reported as mean ± standard deviation of triplicate measurements.

Species of mushroom	Anatomic part	Total phenolic compounds [mg GAE/g d.w.]	
		Water	50% Water–ethanol
	Cultivated		
Agaricus bisporus white	Cap	11.30 ± 0.01	21.23 ± 0.01
	Stipe	11.78 ± 0.03	13.94 ± 0.03
Agaricus bisporus brown	Cap	12.95 ± 0.01	16.93 ± 0.03
	Stipe	14.51 ± 0.01	15.41 ± 0.03
Pleurotus ostreatus	Cap	17.48 ± 0.00	19.33 ± 0.01
	Stipe	13.24 ± 0.01	11.95 ± 0.03
	Wild		
Russula alutacea	Cap	9.30 ± 0.00	22.39 ± 0.10
	Stipe	8.99 ± 0.01	11.66 ± 0.03
Cantharellus cibarius	Cap	7.61 ± 0.03	9.28 ± 0.03
	Stipe	9.41 ± 0.01	13.72 ± 0.00
Russula vesca(m)	Cap	11.13 ± 0.01	34.92 ± 0.33
	Stipe	11.87 ± 0.01	13.72 ± 0.00
Boletus edulis	Cap	31.78 ± 0.03	37.30 ± 0.06
	Stipe	30.97 ± 0.01	13.72 ± 0.00
Agaricus campestris	Cap	17.50 ± 0.00	69.65 ± 0.23
	Stipe	19.94 ± 0.08	14.12 ± 0.00
Macrolepiota procera	Cap	18.32 ± 0.05	22.15 ± 0.11
	Stipe	14.21 ± 0.01	13.85 ± 0.00
Pleurotus ostreatus	Cap	12.67 ± 0.00	15.90 ± 0.01
	Stipe	15.69 ± 0.20	11.68 ± 0.03

determined using water as the solvent. Related to the provenance of the mushrooms, the results from Table 1 show that the group of the cultivated species had lower average values of the polyphenols content than the wild species group, regardless of the solvent used (15.03 mg GAE/g d.w. for water extracts and 20.14 mg GAE/g d.w. for hydroalcoholic extracts). There are no significant differences in the aqueous extracts between caps and stipe both for cultivated and wild species (average value for total phenolic content was 13.54 mg GAE/g d.w. for cultivated and 13.05 mg GAE/g d.w. for wild mushrooms, respectively, except for *B. edulis*). Instead, for 50% water–ethanol extracts, higher values of total phenolic content in cap parts are observed (except for *C. cibarius*) for all species which can suggest a higher content of alcoholic extractible phenolic compounds in caps relative to stipe (Ofodile et al. 2005; Mishra, Pal, and Arunkumar 2014; Mishra, Pal, and Bhatt 2015).

The results from this study are not in agreement with those reported by Boonsong, Klaypradit, and Wilaipun (2016) concerning the antioxidant activities of five edible mushroom species, including two species of *Pleurotus* (*Pleurotus eous* and *Pleurotus sajor-caju*). Thus, among three different solvents (water, 50% (v/v) ethanol, and diethyl ether), the highest amount of total phenolic compounds were found in most water mushroom extracts. However, 50% (v/v) ethanol was found by these authors as the most suitable for antioxidant extraction from the mushroom samples (Boonsong, Klaypradit, and Wilaipun 2016).

The increased amount of the total phenolic compounds in ethanol than in water is also in contrast to the study of Vaskovsky, Khotimcheko, and Boolukh (1998), who reported that more polar constituents are present in mushrooms due to higher solubility with a more polar solvent. The opposite was observed when compared to methanolic extracts of mushrooms. Thus, relatively lower values of phenolic compounds were found in

methanolic extracts of *A. bisporus* by Barros et al. (2008) (4.49 ± 0.1 mg GAE/g extract), Palacios et al. (2011) (3.4 mg GAE/g d.w.), and Ramirez-Anguiano et al. (2007) (4.5 mg GAE/g d.w.), respectively, in ethanolic extracts of the same species by Liu et al. (2013) (6.18 mg GAE/g d.w.).

Sławińska, Radzki, and Kalbarczyk (2013) showed that the concentration of polyphenols in *F. velutipes* was dependent on the solvent and method of extraction. The range of phenolic compounds was the highest in water extracts, next for ethanolic extracts and for acetone extracts, with the methanol extracts lowest.

Also, Reis et al. (2012) reported lower concentration of total phenolic compounds of methanolic extracts for *A. bisporus* brown (37.33 mg GAE/g extract), while *A. bisporus* (white) gave the lowest value (4.22 mg GAE/g extract). However, the results obtained by the above mentioned authors for *P. ostreatus* mushroom in Folin–Ciocalteu assay (12.54 mg GAE/g extract) are quite similar to those in Table 1 for the same species.

Total flavonoid content of mushroom extracts

The total flavonoid content of the 10 analyzed mushrooms (both cap and stipe) is shown in Table 2, the data being calculated as milligrams of quercetin equivalents (QE) per gram of dried weight (d.w.). In relationship with the solvent used, a slightly different situation than in the case of determination of the total phenolic compounds was registered. Thus, there were nine samples (six mushroom caps and three mushroom stipe) from the total of 20, in which the water–ethanol mixture exhibited a better extraction of flavonoids than water. Higher flavonoids were established in the alcoholic extracts of *R. alutacea* (cap), *C. cibarius* (stipe), *R. vesca* (cap) and *P. ostreatus* - wild growing (cap) than in the water extracts.

In the water extracts, flavonoids exhibited concentrations that ranged between 0.22 ± 0.02 mg QE/g d.w. (stipe of *P. ostreatus* cultivated) and 26.51 ± 0.04 mg QE/g d.w. (cap

Table 2. Total flavonoid content of 10 edible mushrooms using two different solvents. Results are reported as mean ± standard deviation of triplicate measurements.

		Total flavonoid content [mg QE/g d.w.]	
Species of mushroom	Anatomic part	Water	50% Water–ethanol
	Cultivated		
Agaricus bisporus white	Cap	6.54 ± 0.00	7.83 ± 4.18
	Stipe	5.23 ± 0.02	3.26 ± 0.04
Agaricus bisporus brown	Cap	8.43 ± 0.02	3.5 ± 0.14
	Stipe	8.05 ± 0.04	5.22 ± 0.06
Pleurotus ostreatus	Cap	2.01 ± 0.02	2.22 ± 0.06
	Stipe	0.22 ± 0.02	1.04 ± 0.04
	Wild		
Russula alutacea	Cap	4.88 ± 0.02	8.97 ± 0.02
	Stipe	2.8 ± 0.02	1.55 ± 0.04
Cantharellus cibarius	Cap	4.62 ± 0.00	1.4 ± 0.15
	Stipe	1.25 ± 0.00	20.77 ± 0.06
Russula vesca(m)	Cap	3.93 ± 0.02	10.69 ± 0.15
	Stipe	3.38 ± 0.02	0.12 ± 0.04
Boletus edulis	Cap	26.51 ± 0.04	10.08 ± 0.04
	Stipe	20.26 ± 0.02	6.37 ± 0.04
Agaricus campestris	Cap	8.31 ± 0.02	10.9 ± 0.10
	Stipe	10.48 ± 0.02	7.8 ± 0.06
Macrolepiota procera	Cap	11.38 ± 0.02	4.72 ± 0.02
	Stipe	5.75 ± 0.00	6.3 ± 0.02
Pleurotus ostreatus	Cap	1.63 ± 0.02	14.58 ± 0.07
	Stipe	10 ± 0.07	3.59 ± 0.06

of *B. edulis),* while in alcoholic extracts, the limits determined were 0.12 ± 0.04 mg QE/g d. w. (stipe of *R. vesca*) and 20.77 ± 0.06 mg QE/g d.w. (stipe of *C. cibarius*). As was described above for the total phenolic compounds, the cultivated species of mushrooms had lower average contents of total flavonoids than the wild ones.

Both for cultivated (excepting alcoholic extract of *A. bisporus brown*) and wild species (excepting aqueous extract of *A. campestris* and alcoholic extracts of *C. cibarius* and *M. procera*), higher flavonoid content was present in the caps for both extracts. According to the results from Table 2, the total flavonoid content varied depending on the mushroom species (Liu et al. 2012; Mishra, Pal, and Arunkumar 2014; Mishra, Pal, and Bhatt 2015) and solvent used, as polar solvents dissolve more flavonoids.

The results related to the flavonoid content were close to those reported by Boonsong, Klaypradit, and Wilaipun (2016) for other two edible mushroom species of *Pleurotus* genera using water as solvent, as follows: *P. eous* 2.61 ± 1.11 mg QE/g d.w. and *P. sajour-caju* 2.29 ± 0.99 mg QE/g d.w. However, the 50% water–ethanol extraction of *Pleurotus* species mentioned above and *Lentinus edodes* was less efficient for flavonoid extraction.

The results obtained in this study were quite different from Palacios et al. (2011) for methanolic extracts of eight types of edible mushrooms, including *A. bisporus, B. edulis, C. cibarius,* and *P. ostreatus*. Thus, smaller values of flavonoid concentrations, ranging between 0.9 and 3.0 mg QE/g d.w., were determined. Also, a previous study by Barros et al. (2008) reported a total concentration of flavonoids expressed as mg of (+)-catechin equivalents (CEs) per gram of methanolic extract of 1.73 ± 0.11 for *A. bisporus*.

Fourier transform infrared spectroscopy

To identify and qualitative confirmation of potential groups involved in antioxidant activity of analyzed mushroom extracts, attenuated total reflection–Fourier transform infrared spectroscopy was used. Fourier transform infrared spectroscopy is an important tool used for quality control and fingerprinting of various range of samples, which allows rapid and simultaneous characterization of different functional groups such as lipids, proteins, nucleic acids and polysaccharides in biological molecules and complex structures (Radulescu 2005).

Fourier transform infrared spectral results of both mushroom cap and stem aqueous and hydroalcoholic extracts, as well as for dried powder blank mushrooms, revealed the occurrence of multiple functional groups. Spectral data for all extract samples confirmed the presence of bioactive functional groups in different concentrations (Tables 3 and 4). All spectra recorded several signals which were attributed to the –OH, >NH, –CHO, –COOH, and –COOR. All extracts exhibited the presence of a broad peak for hydrogen bonded –OH stretching.

For example, the chemical structures of the methanolic extract from *Coriolus versicolor* mycelia biomass have been studied using Fourier transform infrared spectroscopy (Duvnjak et al. 2016). The spectra reported by these authors showed a strong and broadband from OH groups in the spectral region from 3600 to 3000 cm^{-1}. Another study mentioned that this region corresponds to polysaccharides obtained from fruiting bodies of wild basidiomycete *Grifola frondosa* (Klaus et al. 2015). A complete interpretation of the Fourier transform infrared spectra of cap and stem extracts for each type of extract and mushroom species was achieved. The infrared absorption frequencies are summarized in Table 3.

Table 3. Major bands observed in the Fourier transform infrared spectra for aqueous and hydroalcoholic extracts and dried powder blank mushrooms.

Sample	Wavenumber [cm^{-1}]	Relative intensity
Russula alutacea, cap, blank	3268/2922/2849/1637/1584/1457/1372/1315/1260/1196/1147/1078/1019/951/927/876/	w/w/w/w/w/w/w/w/w/w/w/m/s/w/w/w/
Russula alutacea, cap, A	3261/2921/2852/1630/1588/1531/1370/1309/1246/1203/1146/1016/	w/w/w/w/w/w/w/w/w/w/s/
Russula alutacea, cap, E	3268/2936/1626/1589/1372/1317/1258/1199/1148/1077/1019/952/928/883	w/w/w/w/w/w/w/w/w/s/w/w/w/
Russula alutacea, stipe, blank	3264/2935/2851/1639/1586/1454/1374/1320/1260/1196/1147/1081/1020/952/929/884/874/	w/w/w/w/w/w/w/w/w/w/w/m/s/w/m/w/w/
Russula alutacea, stipe, A	3266/2922/1626/1586/1515/1401/1371/1310/1247/1203/1148/1020/894/	w/w/m/w/w/w/w/w/w/w/w/s/w/
Russula alutacea, stipe, E	3252/2935/1622/1583/1451/1375/1322/1260/1195/1146/1080/1019/953/929/888/873/	w/w/w/w/w/w/w/w/w/w/m/s/w/s/w/w/
Cantharellus cibarius, cap, blank	3251/3194/2934/2853/1588/1450/1376/1334/1303/1262/1249/1208/1146/1082/1022/930/890/873/665/	w/w/w/w/s/w/w/w/w/w/w/w/w/s/m/w/w/w/w/
Cantharellus cibarius, cap, A	3269/2920/2852/1631/1586/1377/1309/1243/1025/	w/w/w/w/w/w/w/w/s/
Cantharellus cibarius, cap, E	3250/2921/1588/1377/1306/1147/1023/889/	w/w/s/m/w/w/s/w/
Cantharellus cibarius, stipe, blank	3252/3183/2966/2933/2853/1587/1451/1376/1350/1325/1302/1261/1249/1194/1146/1082/1053/1021/967/930/889/873/	w/w/w/m/w/w/m/w/w/w/w/w/w/w/w/s/w/s/w/s/m/m/
Cantharellus cibarius, stipe, A	3266/2922/2853/1737/1627/1586/1400/1376/1308/1259/1205/1147/1026/	w/m/w/w/m/m/w/w/w/w/w/w/s/
Cantharellus cibarius, stipe, E	3254/2921/2853/1620/1565/1378/1318/1146/1027/	w/w/w/w/w/m/w/w/s/
Russula vesca, cap, blank	3255/3192/2924/2850/1631/1586/1452/1376/1350/1325/1303/1248/1195/1147/1082/1021/967/930/889/873/	w/w/m/w/w/w/w/w/w/w/w/w/w/w/m/s/w/s/w/w/
Russula vesca, cap, A	3279/2923/2853/1736/1635/1581/1451/1382/1302/1250/1184/1048/882	w/w/w/m/s/m/w/w/w/w/w/s/w/
Russula vesca, cap, E	3255/3199/2934/1626/1589/1452/1375/1249/1197/1147/1082/1022/930/889/873/	w/w/w/w/w/w/w/w/w/w/m/s/s/w/w/
Russula vesca, stipe, blank	3254/2936/2359/1633/1589/1453/1375/1352/1322/1261/1196/1147/1081/1021/952/930/888/874/	w/m/w/w/w/w/w/w/w/w/w/w/m/s/w/s/w/w/
Russula vesca, stipe, A	3279/2920/2851/1734/1630/1549/1378/1306/1242/1144/1020/930/885/	w/m/w/w/w/m/w/w/w/w/s/w/w/
Russula vesca, stipe, E	3250/2936/2360/1626/1586/1453/1374/1352/1321/1261/1196/1147/1079/1020/952/929/885/874/	w/m/w/w/w/w/w/w/w/w/w/w/m/s/w/s/w/w/
Boletus edulis, cap, blank	3262/2922/2852/1623/1589/1453/1378/1304/1108/1028/616/	w/w/w/w/w/w/w/w/w/s/w/
Boletus edulis, cap, A	3265/2922/2852/1625/1588/1446/1378/1312/1233/1146/1026/	w/w/w/w/w/w/w/w/w/w/s/
Boletus edulis, cap, E	3264/2922/2853/1623/1590/1377/1306/1234/1105/1028/853/616/	w/w/w/w/w/w/w/w/s/w/w/
Boletus edulis, stipe, blank	3261/2920/1623/1582/1402/1374/1307/1260/1204/1108/1026/686/	w/w/w/w/w/w/w/w/w/s/w/
Boletus edulis, stipe, A	3258/2921/1983/1954/1619/1583/1374/1307/1258/1025/	w/w/w/w/w/w/w/w/w/s/
Boletus edulis, stipe, E	3260/2922/1622/1583/1374/1308/1261/1206/1107/1024/614/	w/w/w/w/w/w/w/w/w/s/w/
Agaricus campestris, cap, blank	3277/2919/2851/1626/1586/1463/1398/1378/1246/1206/1153/1030/720/	w/s/m/w/w/w/w/w/w/w/w/s/w/
Agaricus campestris, cap, A	3266/2920/2851/1627/1584/1453/1396/1311/1231/1151/1029/	w/m/w/w/w/w/w/w/w/w/s/
Agaricus campestris, cap, E	3266/2966/2921/2852/1622/1587/1444/1379/1309/1233/1149/1029/	w/w/w/w/w/w/w/w/w/w/s/
Agaricus campestris, stipe, blank	3261/2921/1626/1581/1374/1309/1203/1149/1026/	w/w/w/w/w/w/w/w/s/
Agaricus campestris, stipe, A	3255/2921/1625/1587/1373/1307/1255/1203/1149/1022/	w/w/m/m/w/w/w/w/w/s/

Agaricus campestris, stipe, E	3263/2924/1623/1589/1373/1306/1235/1207/1150/1025/	w/w/w/w/w/w/w/w/w/s/
Macrolepiota procera, cap, blank	3279/3265/2919/2851/1632/1581/1377/1308/1248/1149/1030/	w/w/m/w/s/s/w/w/w/w/s/
Macrolepiota procera, cap, A	3279/2925/2853/1624/1589/1453/1359/1313/1239/1034/	w/w/w/s/s/w/m/w/m/s/
Macrolepiota procera, cap, E	3265/2923/1625/1587/1376/1309/1237/1149/1028/	w/w/s/s/w/m/w/w/s/
Macrolepiota procera, stipe, blank	3263/2922/1592/1376/1314/1027/777/	w/w/s/w/s/s/w/
Macrolepiota procera, stipe, A	3261/1623/1589/1377/1310/1010/777/	w/w/w/w/w/s/w/
Macrolepiota procera, stipe, E	3252/2923/1621/1594/1375/1308/1012/772/	w/w/w/w/w/m/s/w/
Pleurotus ostreatus wild, cap, blank	3276/2921/2853/1624/1591/1404/1372/1317/1249/1204/1147/1024/994/801/	w/m/w/m/w/w/w/w/w/w/s/w/w/
Pleurotus ostreatus wild, cap, A	3278/2921/2852/1621/1588/1454/1394/1372/1313/1243/1146/1015/927/853/	w/m/w/m/w/w/w/w/w/w/s/w/w/
Pleurotus ostreatus wild, cap, E	3265/2922/1623/1370/1320/1241/1203/1146/1021/993/	w/w/m/w/w/w/w/w/s/w/
Pleurotus ostreatus wild, stipe, blank	3282/2922/2852/1739/1632/1372/1315/1243/1203/1147/1073/1025/994/801/	w/w/w/w/w/w/w/w/w/w/s/w/w/
Pleurotus ostreatus wild, stipe, A	3266/2921/2852/1622/1587/1374/1241/1145/1022/	w/m/w/m/w/w/w/w/s
Pleurotus ostreatus wild, stipe, E	3266/2922/1625/1371/1320/1239/1204/1147/1026/994/	w/w/w/w/w/w/w/s/w/
Agaricus bisporus white, cap, blank	3271/2936/1625/1455/1373/1351/1320/1261/1196/1147/1079/1019/951/928/884/874/	w/w/w/w/w/w/w/w/w/s/s/w/m/w/w/
Agaricus bisporus white, cap, A	3277/2922/2853/1622/1588/1452/1380/1305/1235/1028	w/w/w/m/m/w/w/w/w/s
Agaricus bisporus white, cap, E	3261/2936/1622/1581/1455/1374/1351/1317/1261/1196/1148/1079/1019/951/928/883/875/	w/w/w/w/w/w/w/w/w/w/m/s/w/m/w/w/
Agaricus bisporus white, stipe, blank	3264/2936/1624/1581/1456/1373/1351/1321/1261/1196/1147/1081/1020/952/929/884/874/	w/m/w/w/w/w/w/w/w/w/w/s/s/m/s/w/w/
Agaricus bisporus white, stipe, A	3262/2922/1624/1590/1375/1308/1262/1149/1021/	w/w/w/m/w/w/w/w/s/
Agaricus bisporus white, stipe, E	3261/2936/1623/1588/1455/1375/1351/1320/1261/1196/1148/1079/1020/952/929/884/878/	w/w/w/w/w/m/w/w/w/w/w/s/s/m/s/w/w/
Agaricus bisporus brown, cap, blank	3252/3198/2935/1624/1454/1404/1375/1351/1321/1261/1196/1147/1081/1053/1021/952/930/888/874/	w/w/s/w/m/w/m/w/w/w/w/w/s/w/s/w/s/w/w/
Agaricus bisporus brown, cap, A	3268/2957/2924/2854/1624/1582/1451/1372/1303/1234/1152/1053/1029/	w/w/m/w/s/s/w/m/w/m/w/w/w
Agaricus bisporus brown, cap, E	3262/2936/1622/1588/1455/1374/1351/1261/1197/1147/1079/1020/952/929/884/874/	w/m/w/w/w/w/w/w/w/w/s/s/m/s/w/w/
Agaricus bisporus brown, stipe, blank	3252/3200/2936/1624/1581/1556/1519/1455/1374/1320/1261/1195/1146/1082/1021/952/930/887/874/	w/w/m/w/w/w/w/w/w/w/w/w/w/s/s/w/s/w/w/
Agaricus bisporus brown, stipe, A	3263/2966/2921/2870/2852/1625/1590/1455/1372/1310/1240/1204/1151/1024/1012/883/	w/w/m/w/w/s/s/w/s/m/w/w/w/w/w/w/
Agaricus bisporus brown, stipe, E	3251/2935/1622/1583/1453/1375/1351/1320/1260/1201/1147/1080/1020/952/929/885/874/	w/w/w/w/w/w/w/w/w/w/s/s/w/s/w/w/
Pleurotus ostreatus (cultivated), cap, blank	3260/2924/1582/1517/1454/1403/1147/1027/990/844/801/	w/w/w/w/w/m/w/s/w/w/w/
Pleurotus ostreatus (cultivated), cap, A	3274/2966/2923/2853/1625/1539/1454/1402/1234/1150/1025/930/846/	w/w/m/w/s/w/w/m/w/w/s/w/w/
Pleurotus ostreatus (cultivated), cap, E	3266/2922/1623/1374/1253/1203/1147/1022/989/800/	w/w/w/w/w/w/w/w/w/w/
Pleurotus ostreatus (cultivated), stipe, blank	3271/2925/1624/1371/1203/1148/1073/1023/989/802/	w/w/w/w/w/w/w/m/m/w/
Pleurotus ostreatus (cultivated), stipe, A	3281/2923/1627/1411/1368/1314/1249/1203/1147/1018/895/	w/w/w/w/w/w/w/w/w/s/w/
Pleurotus ostreatus (cultivated), stipe, E	3262/3208/2923/1621/1589/1517/1452/1400/1331/1246/1204/1146/1026/990/841/800/	w/w/w/w/w/w/w/w/w/w/w/w/w/w/w/

A = aqueous extract; E = hydroalcoholic extract; s = strong; m = medium; w = weak.

Table 4. Typical ranges and the main functional groups of identified organic compounds in both mushroom extracts.

Extract	Maximum wavenumber (cm^{-1}) (vibrational mode)	Probable organic compounds
Hydroalcoholic	3260–3266 (H-bonded O—H stretching; broad or –N–H stretching), 3208 (–N–H ring heteroaromatic; mostly in blank samples 3183–3265), 2966 (aldehydic –C–H stretching or carboxylic acid –O–H stretching), 2922–2936 (–C–H stretching), 1638 (–C=C– stretching), 2853 (less in alcoholic extract, only three samples, CH_2–O, ether/HC=O aldehyde), 1621–1623 [aromatic C=C stretch (conjugated), with the presence of C=O], 1583–1592 (–NH bending), 1517 (–NH bending), 1444–1455 (carboxylic acid –OH bending), 1370–1375 (alkane CH_3 bending), 1260–1246 (–C–H in-plane bending), 1197–1203 (amine C—N stretching), 1146–1149 [–C–C(O)–C (acetate) ester stretching], 1081–920 (strong signal: linear/cyclic anhydrides, C–O–Cst), 900–775 (C–H aromatic; o, p, m - subst)	Alkaloids, flavonoids, polyphenols, tannins, carboxylic acids containing phytochemicals, linear/cyclic anhydrides
Aqueous	3255–3281 (H-bonded O—H stretching; broad), 2966 (aldehydic –C–H stretching or carboxylic acid –O–H stretching), 2851–2853 (CH_2-O, ether and HC=O aldehyde), 1734–1736 (aldehyde/ketone –C=O stretching), 1620–1627 [aromatic C=C stretch (conjugated)], 1583–1589 (–NH bending), 1515 (–NH bending), 1446 (carboxylic acid –OH bending); 1368–1374 (–CH_3 bending), 1234–1249 (–C–H in-plane bending), 1203 (amine C—N stretching), 1147–1152 [–C–C(O)–C (acetate) ester stretching], 1018–1025 (linear/cyclic anhydrides, C–O–Cst), 900–775 (C–H aromatic; o, p, m - subst)	Alkaloids, flavonoids, poly phenols, carboxylic acid, cyclic anhydrides

The functional groups of organic compounds from analyzed extracts influence not only their biological activity, but also the acid–base properties, solubility, partition coefficient, crystal structure, and stereochemistry. These properties influence the absorption, distribution, metabolic extraction, and toxicity of bioactive molecules from mushroom extracts. Hence, functional group analysis plays a vital role in understanding the overall physicochemical properties of the extract. Also, the identification of the functional groups helps to evaluate their structure–activity relationships.

In the present study, the Fourier transform infrared spectra of the mushroom extracts demonstrate the presence of different organic compounds performing hydrogen bonded –OH functional groups which are involved in antioxidant and anti-inflammatory activities. The results indicate that the hydroalcoholic mushroom extracts contain various biologically active functional groups, such as alcohols, esters, and aldehydes.

Statistical analysis

Pearson correlation analysis (significance level $p < 0.01$; confidence level of 99%) was applied to evaluate the influence of the total phenolic compounds and flavonoid content of mushrooms on their antioxidant activity in relation with the solvent used. The use of water as the solvent leads to a lack of correlation between the antioxidant activity and phenolic compound ($r = -0.133$) and flavonoid content ($r = -0.084$) of mushrooms. The results from bivariate analysis indicate that there are other factors that influence the antioxidant activity. Moreover, the parameters are inversely correlated, as it is obvious that the

method of extraction needs to be reviewed in this context. A firm correlation between the phenolic compounds and flavonoids using water as solvent was established ($r = 0.885$).

In the case of the hydroalcoholic samples, a direct and high correlation ($r = 0.717$) it is observed between the phenolic compounds and antioxidant activity of mushrooms (both caps and stipe), while a relative weak relationship ($r = 0.426$) is present between the flavonoid content and antioxidant activity.

Factor analysis is a multidimensional statistic method applied for data reduction, for highlighting and fixing of the pattern of correlations between variables and also for determination of the fewer latent variables found behind the measured variables. The method is similar to the principal component analysis, as the factor weights are scaled.

Applying factor analysis to the experimental results, which were included all mushrooms species (caps and stipe), the solvent used (water and 50% water–ethanol), the phenolics, the flavonoids, and the antioxidant activity, the analytical variables were reduced to two principal components, which accounted for 40.95% (principal component 1) and 33.21% (principal component 2) of the total variance. According to the component matrix, the most notable variable was the antioxidant activity of the mushroom hydroalcoholic extracts. To get a better angle of view, rotation by the Varimax method was applied (Figure 2).

The combination of principal component 1 (PC1) and principal component 2 (PC2) underlines the lack of correlation between the antioxidant activity and total phenolic compounds and the flavonoids for the aqueous extracts. In the same time, it is obvious that principal component 1 affected the independent variables (flavonoids and phenolic compounds) when water was used as solvent, while principal component 2 mostly separated the variables correlated with ethanol as the solvent.

If the results related to the mushroom species are subjected to factor analysis in a differentiated way in relationship with the type of the solvent, the analytical variables are reduced to two principal components (Figure 3) for the alcoholic extracts. Only one

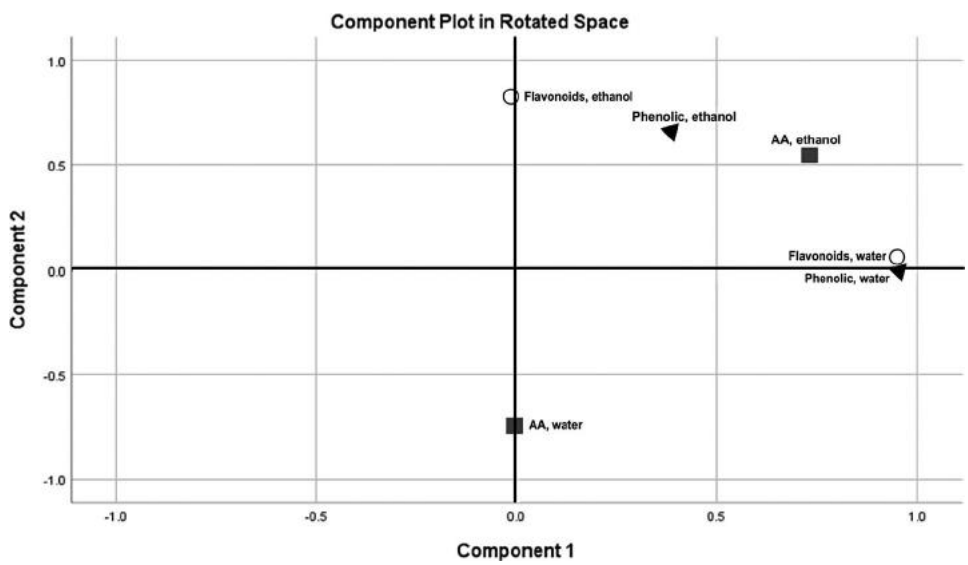

Figure 2. Component plot in rotated space.

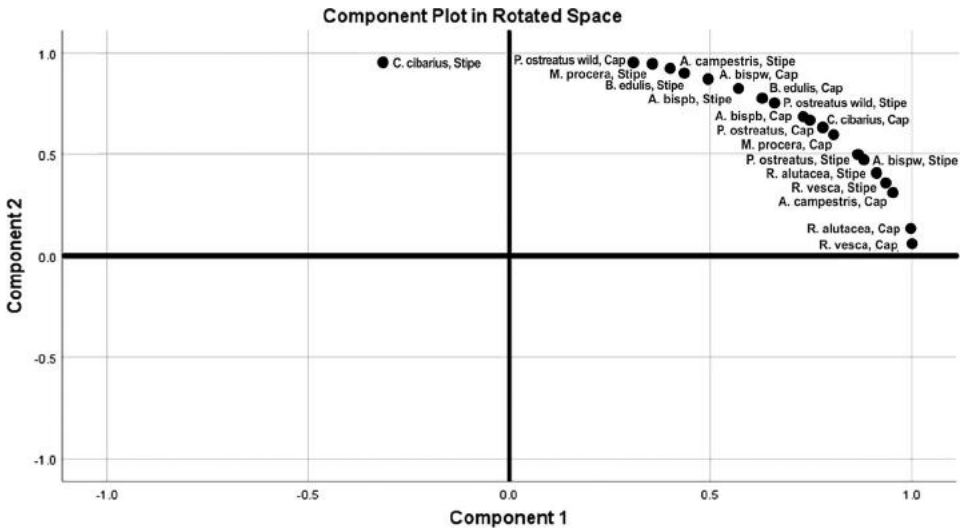

Figure 3. Component plot in rotated space (hydroalcoholic extracts of the mushroom species).

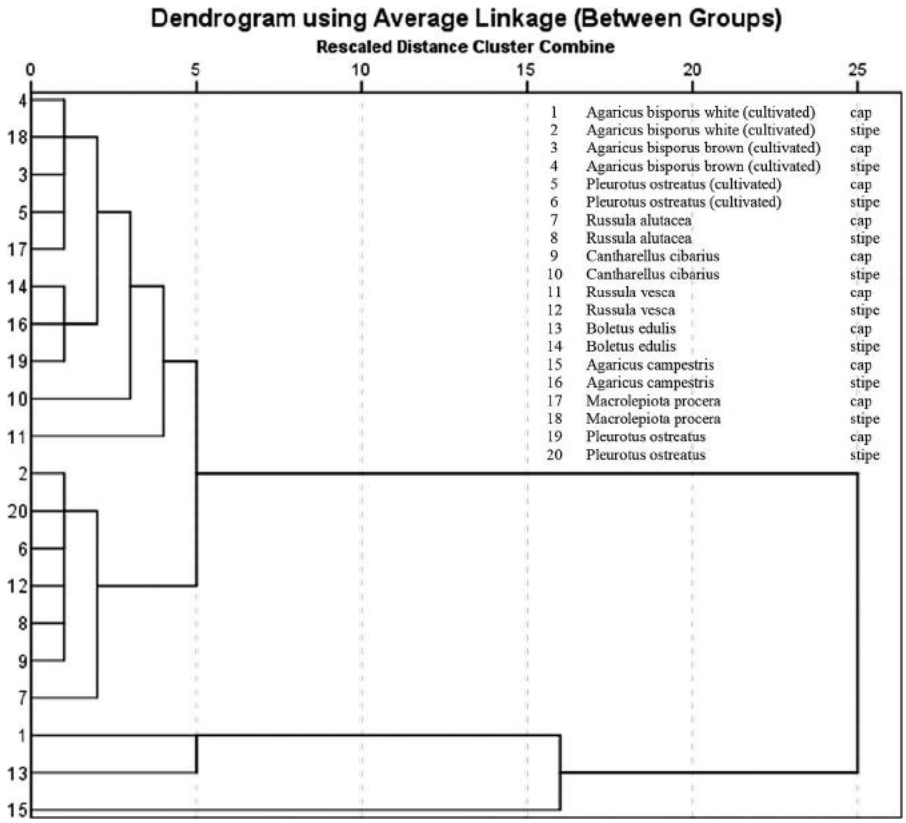

Figure 4. Hierarchical cluster of the hydroalcoholic extracts.

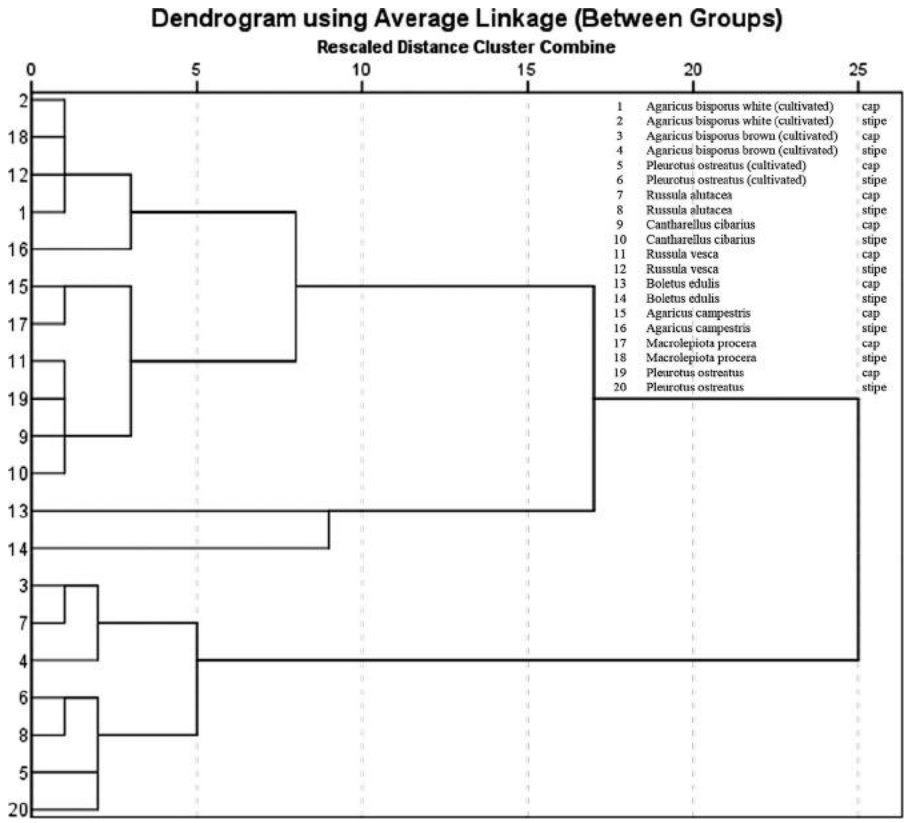

Figure 5. Hierarchical cluster of the aqueous extracts.

component was extracted in the case of analysis of the set of variables that characterized the aqueous samples.

Hierarchical cluster analysis was applied to identify the grouping after the reduction of the dimensions and implicitly few variables through factor analysis which also minimizes multicollinearity effects. The dataset related to the 50% water–ethanol extracts was joined into clusters, thus increasing their in-group homogeneity (Figure 4). The differences among the clusters and the individual groups are obvious in the case of the dataset related to the aqueous extracts (Figure 5). The smallest squared Euclidean distances between cases were observed in the frame of the group represented by the samples extracted with water–ethanol mixture (Figure 4). It cannot be stated that the anatomical part of the mushrooms—as an independent variable—had a decisive influence in grouping of the data and clustering.

Conclusion

This is the first study that characterized traditional mushroom species by antioxidant activity correlated with their total content of phenolics and flavonoids. Due to the smaller differences related to the chemical constituents and the 2,2-diphenyl-1-picrylhydrazyl radical-scavenging ability between the cap and stipe of mushrooms, separating these parts

for processing and consumption of stipe should be avoided. The water extracts of *B. edulis* exhibited the highest amounts of polyphenols and flavonoids than the other species, while its hydroalcoholic extract showed the highest antioxidant activity. For these reasons, *B. edulis* could be used as potential natural source of antioxidant additives. Higher and also average values for all the evaluated parameters were determined for the wild growing mushrooms.

Additional studies on the mushroom extracts (including the expanding of the type of solvents) are needed to obtain, as much as possible, a more accurate overview on the secondary metabolites responsible for their antioxidant activity. The statistical approach highlights a clearly link between phenolic compounds and antioxidant activity of mushroom extracts (both caps and stipe) depending on solvent type. This study represents a novel starting point for future studies in which these or other extracts can be used in various fields such as medicine, pharmaceutics, cosmetics, and geriatrics.

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